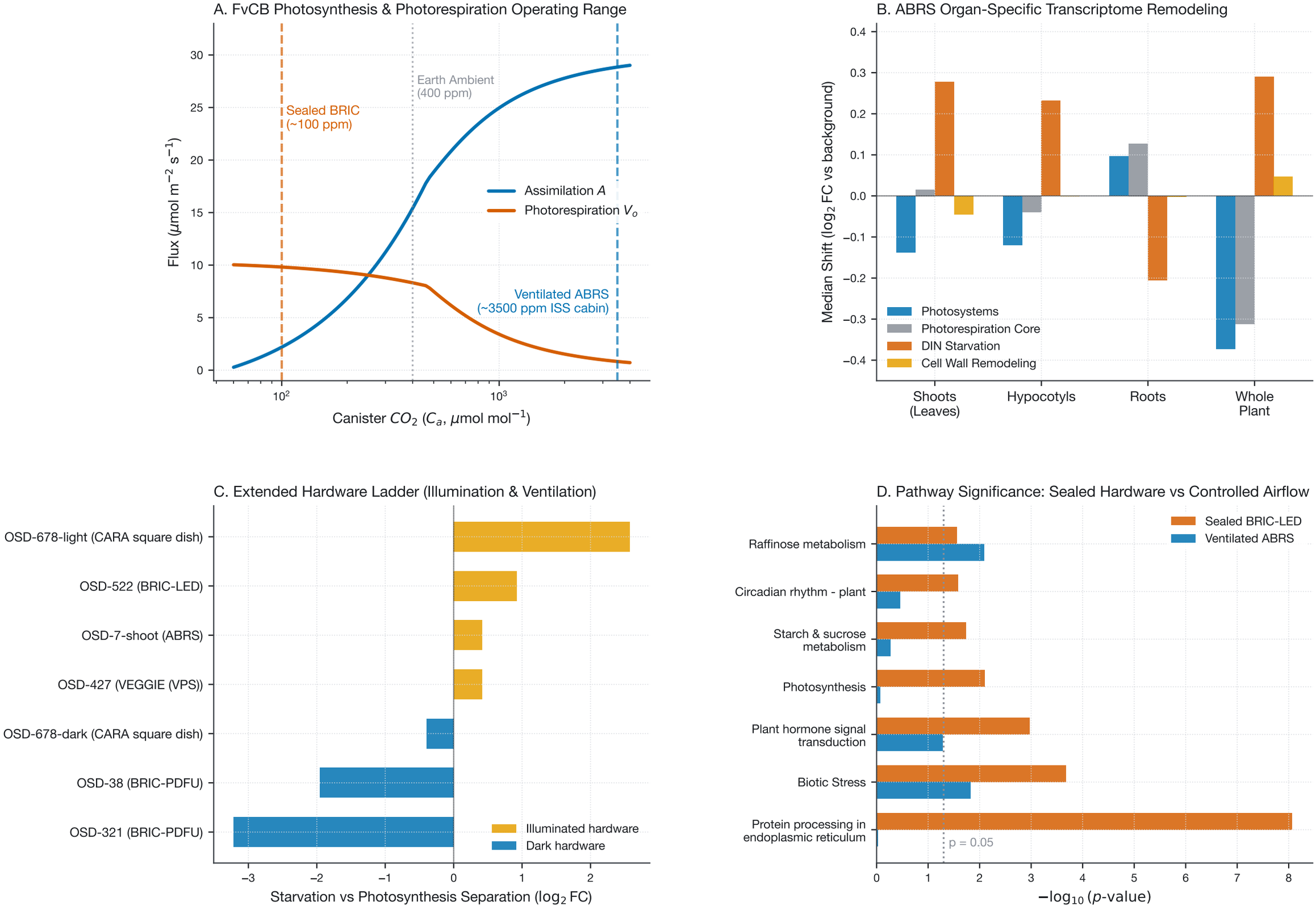
A. FvCB Photosynthesis & Photorespiration Operating Range
Flux (μmol m⁻² s⁻¹)
Canister CO2 (Ca, μmol mol⁻¹)
Sealed BRIC
(~100 ppm)
Earth Ambient
(400 ppm)
Ventilated ABRS
(~3500 ppm ISS cabin)
Assimilation A
Photorespiration Vo
B. ABRS Organ-Specific Transcriptome Remodeling
Median Shift (log2 FC vs background)
Photosystems
Photorespiration Core
DIN Starvation
Cell Wall Remodeling
Shoots (Leaves)
Hypocotyls
Roots
Whole Plant
C. Extended Hardware Ladder (Illumination & Ventilation)
OSD-678-light (CARA square dish)
OSD-522 (BRIC-LED)
OSD-7-shoot (ABRS)
OSD-427 (VEGGIE (VPS))
OSD-678-dark (CARA square dish)
OSD-38 (BRIC-PDFU)
OSD-321 (BRIC-PDFU)
Illuminated hardware
Dark hardware
Starvation vs Photosynthesis Separation (log2 FC)
D. Pathway Significance: Sealed Hardware vs Controlled Airflow
Sealed BRIC-LED
Ventilated ABRS
Raffinose metabolism
Circadian rhythm - plant
Starch & sucrose metabolism
Photosynthesis
Plant hormone signal transduction
Biotic Stress
Protein processing in endoplasmic reticulum
p = 0.05
−log10 (p-value)